Structural Boolean circuits and subexponential sample bounds for the original Gaussian tanh posterior

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Abstract

We give structural sufficient conditions for subexponential dependence on learner depth in the original, unrestricted Gaussian tanh posterior. A synchronous Boolean circuit with an early narrow layer can be evaluated by retaining its prefix and replacing its suffix by a truth table. A bounded-fanin circuit of small graph treewidth admits a shallow threshold implementation by tree-decomposition dynamic programming. For planar circuit graphs, a separator schedule improves the exponent to the square root of graph size, apart from logarithmic terms. We account for the size and coefficients of these constructions, and give the resulting sample bounds at every realizable learner width, uniformly over canonical Gaussian scales $0 < t \le 1$. The statistical step imports the established uniform intrinsic-learning and scalar-continuation theorems; a heterogeneous-layer prefix lemma needed here is proved explicitly. These results do not cover arbitrary polynomial-size deep Boolean circuits, nor does it replace real tanh states by their signs.

1 Model and statements

Let $X_d = \{-1,1\}^d$, $f : X_d \to \{-1,1\}$, and $H = L - 1 \ge 1$. The learner has H hidden tanh layers of common width N, input $(1, x)$, and a scalar linear readout. All weights are independent centered Gaussians, with entry variances

$$\mathrm{Var}(W^1_{ij}) = t^2/d, \qquad \mathrm{Var}(W^j_{ik}) = t^2/N \ (j \ge 2), \qquad \mathrm{Var}(v_i) = t^2/N^2, \quad 0 < t \le 1.$$

There are no additional hidden biases. A constant input coordinate may be propagated as an ordinary neuron. Write $z_w(x)$ for the output and

$$A_f = \{w : f(x) z_w(x) > 1/2 \text{ for every } x \in X_d\}, \qquad a_{N,t}(f) = -N^{-1} \log \mu_{N,t}(A_f).$$

A width is *realizable* if $\mu_{N,t}(A_f) > 0$. The posterior conditions on $|z_w(x)| > 1/2$ at every cube input and on agreement with all noiseless training labels. Training and test inputs have any common distribution $\mathcal{D}$. Its confidence error is

$$\mathcal{D}\{x : \Pr_{\text{post}}(\operatorname{sign} z_w(x) = f(x)) \le 2/3\}.$$

Every statistical assertion below is at each fixed realizable width; it does not assert one simultaneous event for infinitely many widths.

For $E \ge 1$, abbreviate the sufficient sample bound

$$n \ge \frac{C}{\varepsilon}\left[e^{CE}(1 + \log(1/\varepsilon))^C + \log(1/\delta)\right], \qquad 0 < \varepsilon, \delta < 1. \tag{1}$$

Here and below C denotes a universal constant that may increase.

Theorem 1.1 (An early narrow cut)**.** *Suppose f is computed by a synchronous fan-in-two Boolean circuit with $s \geq 1$ gates and depth $h \geq 1$: inputs occur at level zero, and every wire joins adjacent levels. Let w_j be the number of bits at level j, $1 \leq j \leq h$. Put*

$$K = \log(d + s + L + 3), \qquad g = \frac{K}{\log(K+2)}, \qquad E_{\mathrm{cut}} = K + \min_{1 \leq j \leq h} (jg + w_j).$$

If $L \geq CE_{\mathrm{cut}}$, then (1) with $E = E_{\mathrm{cut}}$ bounds the confidence error by ε with training probability at least $1 - \delta$, at every realizable width and every $0 < t \leq 1$. In particular one may replace E_{cut} by

$$E_{\mathrm{size}} = K + \sqrt{sK/\log(K+2)}. \tag{2}$$

If inputs are also injected after a cut, the narrow-cut assertion holds with w_j replaced by $w_j + r_j$, where r_j counts the distinct input bits used strictly after that cut. The size-only bound (2) requires the original synchronous, level-zero-input hypothesis.

Theorem 1.2 (Small treewidth)**.** *Suppose a fan-in-two Boolean circuit computing f has $v \geq 1$ vertices, including its input vertices, and its underlying undirected graph has treewidth k. Set*

$$E_{\mathrm{tw}} = \log(d + L + 3) + (k + \log(v + d + L + 3)) \log(v + 3).$$

If $L \geq CE_{\mathrm{tw}}$, then (1) with $E = E_{\mathrm{tw}}$ has the same confidence-error, scale, and all-width conclusions.

Theorem 1.3 (Planar and excluded-minor circuit graphs)**.** *Suppose the underlying undirected graph of a fan-in-two Boolean circuit for f is planar, with $v \geq 1$ vertices including its input vertices. Set*

$$E_{\mathrm{sep}} = \sqrt{v} + \log(d + L + 3) + \log(v + 3) \log(v + d + L + 3).$$

If $L \geq CE_{\mathrm{sep}}$, then (1) with $E = E_{\mathrm{sep}}$ has the same confidence-error, scale, and all-width conclusions. The same statement holds for any fixed excluded-minor graph class, with constants depending only on the excluded minor.

For families with $d, s, v \leq L^{O(1)}$, these theorems give an $\exp(o(L))$ sample bound at fixed ε, δ whenever, respectively,

$$\min_j (j \log L / \log\log L + w_j) = o(L), \quad s = o(L^2 \log\log L / \log L), \quad \text{or} \quad k = o(L/\log L).$$

The planar or fixed-excluded-minor bound is subexponential when $v = o(L^2)$. The displayed padding conditions then hold eventually. In particular, they must not be omitted from finite-parameter versions of the results. The conclusions restrict the *target's* Boolean representation; they impose no sparsity, treewidth, or architecture restriction on the prior or posterior.

2 The analytic learning inputs and a layerwise prefix

This section isolates exactly which prior analytic results are used. Let

$$b = t^{-2} - 1, \qquad c_t = (t^{-2} - 1 + 2\log t)/2,$$

and let V_* denote the critical scalar Bellman baseline of [2, 1]. Its numerical value is not needed.

Proposition 2.1 (Imported intrinsic and scalar inputs)**.** *The following statements hold for the model above.*

1. *For the actual target cost $a = a_{N,t}(f)$, or a deterministic upper bound for it, define*

$$E_a = (a + 4/(Ht^2) - Hc_t - V_* + b)_+, \qquad J_a = \min\{\sqrt{HE_a}, E_a/b\},$$

omitting the second term when $b = 0$. A sufficient sample size is polynomial in $d, L, 1 + t^2 a$, multiplied by $\exp(CJ_a)$ and the statistical factors in (1). The polynomial degrees and constants are universal, and the conclusion holds at every realizable width uniformly in $0 < t \le 1$.

2. *Suppose a Boolean circuit prefix of q hidden stages supplies a target component $p \ge e^{-CS}$, diagonal at most one, critical cost at most $1/(8L)$, and finite-width relative score and residual accuracy $e^{-C'S}$ at widths $N \ge e^{C''S}$, for any required fixed exponents after increasing constants. Suppose $S \ge 1 + \log(d + L + 2) + q$. Its Gaussian mean shifts at scale t have cost at most $1/(8Lt^2)$. If $H - q \ge CS$, the critical scalar program, reused at scale t, gives the actual target-mass bound*

$$a_{N,t}(f) \le Hc_t + V_* + C(bS + S^2/H) \quad (N \ge e^{CS}), \tag{3}$$

with constants and width threshold independent of t.

The first item is the baseline-subtracted intrinsic bound in [1, Proposition 2.1, equation (4)], proved in the companion note `uniform-sharp-intrinsic.md`. The second is the scalar continuation and finite-width lift in [1, Proposition 3.3] (label `prop:long`), with the prefix supplied separately. Its proof uses the prefix only through the listed seed, cost, and accuracy bounds. The underlying finite-width circuit prefix is [2, Lemma `lem:critical-finite-circuit-prefix`]. These are imported analytic theorems, not assertions reproved by the combinatorial arguments below.

Lemma 2.2 (A heterogeneous-layer prefix)**.** *Consider a layered threshold circuit with $q \ge 1$ stages, widths $C_j \ge 1$, a constant rail, unit normalized gate margins, and row Euclidean norms at most $B_j \ge 1$ at stage j. Set*

$$S = 1 + \log(d + L + 3) + q \log(q + L + 3) + \sum_{j=1}^{q} \log((C_j + 1)(B_j + 1)). \tag{4}$$

For $L \ge q+1$ this circuit supplies the prefix hypotheses in Proposition 2.1, with this S. Consequently, when $L \ge CS$, (1) holds with $E = S$ uniformly over all canonical scales and all realizable widths.

Proof. We give the resource dependence, including the finite-width construction, to avoid replacing the sum in (4) by a maximum width times depth. Put $p_0 = 1/d$, and choose a common gain a and tolerance $\eta = 8e^{4-a}$. Define

$$p_j = \frac{p_{j-1}}{8C_j q L a^2 B_j^2}, \qquad k_j = \lceil N p_j \rceil.$$

For each wire prescribe a disjoint group of k_j neurons. In the first layer use mean $aw_{1,i}$ on a controlled row. At stage $j > 1$, put mean $aw_{j,i,g}/k_{j-1}$ on each coordinate in the preceding group for wire g, and zero elsewhere. All row covariances retain their original Gaussian values. The preactivation mean is exactly $a \sum_g w_{j,i,g} m_{j-1,g}(x)$, where $m_{j-1,g}$ is that group's empirical mean.

If $Np_q \ge 1$, then $k_j \le 2Np_j$, the groups fit, and the entropy per controlled row is at most $a^2 B_j^2/(2t^2 p_{j-1})$. Thus

$$\frac{1}{N} \mathrm{D}(P \| \mu_{N,t}^{\mathrm{prefix}}) \le \sum_{j=1}^{q} \frac{2C_j p_j a^2 B_j^2}{2t^2 p_{j-1}} \le \frac{1}{8Lt^2}.$$

Choose $\eta \le (4\max_j B_j\sqrt{W_j})^{-1}$, where $W_1 = d+1$ and $W_j = C_{j-1}$ thereafter. An incoming empirical error at most η then leaves preactivation mean of the correct sign and magnitude at least $3a/4$. Gaussian variance is at most two at the first stage and one thereafter. Gaussian exponential moments and $1-\tanh u \le 2e^{-2u}$ show that its tanh expectation is within $\eta/4$ of the correct bit, and its second moment is within $\eta/2$ of one.

Conditional Hoeffding bounds, united over all cube inputs, groups, and stages, now give simultaneous gate-mean error at most η provided

$$Np_q \ge C\eta^{-2}\left[d + \log\frac{q(1+\max_j C_j)}{\zeta}\right].$$

The diagonal follows the scalar recurrence $\bar{q}_j = (1-\theta_j)F_t(\bar{q}_{j-1}) + \theta_j$, where $\theta_j = C_j k_j/N$, $\bar{q}_0 = (d+1)/d$, and $F_t(u) = \mathbb{E}\tanh^2(t\sqrt{u}G)$. Since F_t is one-Lipschitz for $t \le 1$, bounded-square concentration gives diagonal error at most $2j\eta$. The final target-group indicator, normalized to a unit vector, therefore has score within $\sqrt{p}\eta$ of $\sqrt{p}f$, where $p = k_q/N$, and residual-variance error at most $(2q+3)\eta$. After the first stage the ideal diagonal is at most one. This is the same score/residual seed used by the scalar continuation.

Take $a = AS$ for a sufficiently large universal A. Because $S \ge q\log(q+L+3)$ and $S \ge q$, one has

$$q\log S = q\log q + q\log(S/q) \le 2S.$$

It follows that

$$\log(1/p_q) = \log d + \sum_j \log(8C_j q L a^2 B_j^2) \le C(1+\log A)S.$$

The gate-margin tolerance and any fixed seed-relative accuracy requirement hold by increasing A: $\log(1/\eta)$ grows linearly in A, while the last bound grows only logarithmically in A. The concentration requirement is then $N \ge e^{CS}$ for $\zeta \ge e^{-C'S}$, with arbitrary fixed C'. The construction and all concentration estimates are uniform for $t \le 1$.

For $L \ge CS$, Proposition 2.1 gives (3). In its intrinsic bound, $E_a \le C(bS + S^2/H)$, absorbing b and $4/(Ht^2)$ because $S \ge 1 + \log(H+2)$. If $b \ge S/H$ use E_a/b; otherwise use $\sqrt{HE_a}$. In either case $J_a = O(S)$. Also $t^2 a \le C(H + S + S^2/H)$, so all polynomial factors are absorbed by e^{CS}.

This proves the conclusion for $N \ge e^{CS}$. For smaller widths, the entire depth-L, width-N tanh sign class has VC dimension polynomial in its parameter and computation counts [7]. These counts are $e^{O(S)}$, since $d, L, N \le e^{O(S)}$. The standard realizable VC consistency bound therefore makes every posterior-supported function's test error at most $\varepsilon/3$ under (1). Averaging over that posterior and applying Markov's inequality gives confidence error at most ε: at a point counted as an error, posterior mistake probability is at least $1/3$. This handles every smaller realizable width without requiring the compiled circuit to fit inside that width. □

3 A prefix followed by one lookup

We first record the resource accounting behind cone blocking. Take b consecutive fan-in-two Boolean levels. Each output depends on at most 2^b entrance bits. Its truth table has a DNF with at most 2^{2^b} terms, hence two threshold stages, whose widths and normalized row norms have logarithms $O(2^b + \log(s+2))$. Copy gates and a constant rail obey the same bounds. Thus a depth-j prefix has a threshold implementation with

$$q_0 = O(1+j/b), \qquad \sum_{\text{prefix}} \log((C_i+1)(B_i+1)) = O((1+j/b)(K+2^b)).$$

The same bounds hold if finitely many specified input bits must be retained for later use. Choose b within a constant factor of $\log(K+2)$, truncated to $1 \le b \le j$. Then $2^b = O(K)$ and

$$S_{\text{prefix}} = O(K + jK/\log(K+2)). \tag{5}$$

This is also the resource-sensitive blocking construction in [2, `thm:critical-boolean-blocking`].

At a cut with w bits, the whole suffix is a Boolean function of those bits. Its DNF implementation has two threshold stages, width $2^{O(w)}$, and normalized row norms $2^{O(w)}$. It contributes $O(w+1)$ to the sum in (4), not $O(q_0 w)$. Combining with (5) gives $S = O(K + jg + w)$ and proves the first part of Theorem 1.1 by Lemma 2.2. If some input bits enter after the cut, retain precisely those distinct bits alongside the cut state. This proves the asserted $w + r$ correction.

For completeness, first prune all gates that are not ancestors of the designated output; this can only reduce the widths and gate count. Take the first remaining level with at most y gates. It exists because the final output is one bit. All previous levels have more than y gates, so its index is at most $s/y + 1$. Therefore

$$E_{\text{cut}} \le C\{K + y + (s/y)g\}.$$

Choosing a positive integer y within one of $\sqrt{sg}$ gives (2). This keeps the prefix of the synchronous depth-reduction argument of Gál–Jang [3], instead of using only its minimum-depth conclusion and losing size control.

Remark 3.1 (Ordinary neural realizations). The same threshold implementations have unit-margin tanh realizations with logarithmic Frobenius product bounded by their parameter S. Indeed, use a common output-bit tolerance $\eta = e^{-CS}$ and tanh gain $a = O(S)$ at each threshold stage. Gate scores retain a fixed sign gap, and saturation achieves that tolerance. The layer norm is at most $aB_j\sqrt{C_j}$; hence

$$\log \prod_j \max\{1, \|W_j\|_F\} \le q \log a + \sum_j \log B_j + \tfrac{1}{2} \sum_j \log C_j = O(S).$$

A final scalar multiplier gives unit output margin. This product statement is additional evidence for the resource accounting; the posterior conclusion above uses the Gaussian seed construction and does not assume a generic product-to-target-mass theorem.

4 Tree-decomposition dynamic programming

Start with a tree decomposition of the underlying circuit graph of width k. The balanced-decomposition theorem gives a binary decomposition of width at most $3k+2$, depth $O(\log(v+3))$, and polynomially many bags [4, Lemma 1]. This external graph theorem is the only structural input in this section.

Enlarge each bag by adding the at most two predecessors of every circuit vertex already in it. Its size grows by a factor at most three. This remains a tree decomposition: for any vertex u, its new bags are its old connected bag subtree together with the old subtrees of its successors. Each such subtree intersects the old subtree of u, because a bag covers that circuit edge. Their union is connected. Moreover each gate's full constraint scope—the gate and its at most two inputs—is now contained in a bag.

Introduce one Boolean variable z_u for every circuit vertex. For a fixed external input x, require every gate equation, every input equation $z_u = x_i$, and $z_{\text{out}} = 1$. Anchor each constraint at one bag containing its variables. Because the circuit is acyclic, the whole system is satisfiable exactly when the circuit accepts x.

For bag B_t and assignment $a \in \{0,1\}^{B_t}$, let $A_t(a;x)$ assert that a extends to a satisfying assignment of all constraints anchored below t. If t has children ℓ, r, then

$$A_t(a;x) = \mathrm{Local}_t(a;x) \wedge \bigvee_{b:\, b|_{B_t\cap B_\ell}=a|_{B_t\cap B_\ell}} A_\ell(b;x) \wedge \bigvee_{c:\, c|_{B_t\cap B_r}=a|_{B_t\cap B_r}} A_r(c;x). \tag{6}$$

For a missing child omit its factor. Running intersection shows that the two child subtrees overlap only in variables already fixed at the parent; hence compatible extensions glue. This proves both directions of (6). For fixed a, the anchored gate constraints and output condition are constants, while the input constraints are a conjunction of at most $O(k+1)$ literals of x.

Compute all these local terms first. Retain them by copy gates until their merge level, and evaluate (6) upward. Each merge uses an OR stage and an AND stage. Finally OR over all root assignments. There are $2^{O(k)}\,\mathrm{poly}(v+d)$ table entries and connections, and $O(\log(v+3))$ threshold stages. All thresholds can be chosen integral and normalized to margin one; their row norms are at most $2^{O(k)}\,\mathrm{poly}(v+d)$. Constant rails, copying pending tables, and retaining local terms obey the same resource bounds. In particular the potentially exponential table size is charged explicitly.

Substituting these resources into (4) gives

$$S \le C\{\log(d+L+3) + (k+\log(v+d+L+3))\log(v+3)\} = O(E_{\mathrm{tw}}).$$

Lemma 2.2 proves Theorem 1.2. The preceding remark also gives a constant-margin tanh realization of physical depth $O(\log(v+3))$ and logarithmic Frobenius product $O((k+\log(v+d+3))\log(v+3))$, choosing its gain without the statistical learner-depth parameter.

5 Geometrically shrinking separators

The weighted planar separator theorem [5] partitions any n-vertex planar graph with nonnegative vertex weights into two nonadjacent sides and a separator of at most $c\sqrt{n}$ vertices, each side carrying at most $\beta = 2/3$ of the total weight. We use this theorem twice at each recursion step, for two different weights.

A subproblem consists of disjoint interior and boundary sets I, B, with the induced graph on $I \cup B$; no interior vertex has an edge outside this subproblem. Initially I is the whole graph and $B = \varnothing$. Write $m = |I|$, $b = |B|$. If $I = \varnothing$, use the leaf bag B and stop. Otherwise, first separate with weight one on I and zero on B, obtaining separator Z_0 and two sides U_0, U_1, each with at most βm interior vertices. Within each nonempty U_i, apply the weighted theorem with weight one on $B \cap U_i$ and zero elsewhere. If this weight is zero, retain U_i as one side without a separator. Let Z_1, Z_2 be the additional separators and P_a the at most four resulting sides. Put

$$Z = Z_0 \cup Z_1 \cup Z_2, \qquad \mathcal{B}_t = B \cup Z, \qquad I_a = I \cap P_a, \quad B_a = (B \cap P_a) \cup Z.$$

The children are the induced subproblems on $I_a \cup B_a$. Duplicating all of Z into each child is allowed. Their sizes obey

$$|I_a| \le \beta m, \qquad |B_a| \le \beta b + 3c\sqrt{m+b}. \tag{7}$$

Different P_a have no edges between them, so the boundary invariant persists. The bags $\mathcal{B}_t$ form a tree decomposition: edges are either covered in a parent bag or passed to one child, and a vertex

appearing in several children belongs to the parent bag. Induction gives edge coverage and running intersection.

Put $M_\ell = \beta^\ell v$. Choose a universal A large enough that

$$\beta A + 3c\sqrt{1+A} \le A\sqrt{\beta};$$

such an A exists because $\beta < \sqrt{\beta}$. At depth ℓ, interior size is at most M_ℓ and boundary size is at most $A\sqrt{M_\ell}$. Indeed, when $M_\ell \ge 1$, (7) and this induction hypothesis give

$$\begin{aligned} |B_a| &\le \beta A\sqrt{M_\ell} + 3c\sqrt{M_\ell + A\sqrt{M_\ell}} \\ &\le (\beta A + 3c\sqrt{1+A})\sqrt{M_\ell} \le A\sqrt{M_{\ell+1}}. \end{aligned}$$

When $M_\ell < 1$, interiors are empty and recursion has stopped. Thus the decomposition has depth $O(\log(v+3))$, polynomially many bags, and every bag at depth ℓ has size $O(\sqrt{M_\ell})$. The last level has constant bag size. Enlarge bags by gate predecessors exactly as in the preceding section. Their sizes increase by at most three, while all gate scopes become covered; no planarity of the enlarged scope graph is needed.

Now perform the same bag-assignment dynamic program, scheduling levels from deepest to root. Each table entry is a local term AND at most four compatible-child ORs, so each level uses only a constant number of threshold stages. Carry the original Boolean input bits and a constant rail throughout, and compute each bag's local literal constraints only when that bag's level is reached. Leaf tables are also created at their own level. Child tables are discarded after the next merge. In particular, large root tables are not created at the first physical layer and retained throughout; doing so would lose the following improvement.

At level ℓ, logarithmic widths and normalized row norms are

$$O(\sqrt{M_\ell} + \log(v+d+3)).$$

This includes input rails, local terms, all tables at that level, and their connections. The geometric series satisfies

$$\sum_{\ell \ge 0} \sqrt{M_\ell} \le \frac{\sqrt{v}}{1-\sqrt{\beta}}.$$

With $q = O(\log(v+3))$ physical threshold stages, the sum in (4), including its $q\log(q+L+3)$ term, is therefore $O(E_{\text{sep}})$. Lemma 2.2 proves the planar case of Theorem 1.3.

For a fixed excluded minor, use the weighted $O(\sqrt{n})$ separator theorem of Alon–Seymour–Thomas [6]; the same argument applies with its fixed constant. Its component balance can be grouped into two sides of weight at most 2/3. This proves the extension.

6 Scope of the conclusions

These are positive target classes for the original posterior, including its unrestricted background neurons and all canonical Gaussian scales. Small compiled width is unnecessary at an individual learner width: at the smaller widths, the proof uses the full finite-width class rather than forcing the compilation into the learner. Realizability remains necessary for conditioning on noiseless target labels.

The bounds do not solve the generic Boolean-depth question. A polynomial-size circuit can have large treewidth and no early sublinear-bit cut, and need not lie in a fixed excluded-minor

class. Generic planarization can have quadratic size cost, which removes the sublinear-in-size gain of Theorem 1.3. Size-only depth reductions need not preserve the resource sum (4). Moreover a bounded-width tanh teacher stores real-valued states; its hidden signs need not be sufficient statistics for the next layer. Applying these results to such a teacher requires an independently justified Boolean simulation with the stated structural resources. No cutset or treewidth limitation of one representation is an intrinsic neural lower bound, and no exponential sample lower bound is proved here.

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